



Does centrality within trade agreements network matter to economic complexity? The conditioning effects of network structure

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ABSTRACT

This paper investigates how a country's centrality within an ego-centred trade agreements network impacts its economic complexity, emphasizing the conditioning effects of network structure. We propose that enhancing a country's centrality yields both a "breadth effect" and a "depth effect" on its production capabilities by facilitating the transnational diffusion of knowledge. This diffusion positively impacts the country's economic complexity. Furthermore, we contend that a country's centrality within a "non-natural" trade agreements network—comprising countries with diverse production capabilities—has a stronger influence on its economic complexity. Additionally, we argue that developed countries predominantly serve as major sources for transnational knowledge diffusion, thereby enhancing the impact of a country's centrality within the "developed" trade agreements network on its economic complexity. Our empirical analysis, based on panel data for 78 countries over the 2000–2019 period, supports these hypotheses. The findings are validated through robustness tests, including an examination of potential endogeneity issues. This paper contributes to the literature on policy-related determinants of economic complexity by highlighting the pivotal role of strategic positioning within the trade agreements network.

1. Introduction

In the past few decades, there has been a significant proliferation of preferential trade agreements (PTAs) worldwide. Between 1990 and 2022, the number of such agreements surged from 70 to 570, indicating a substantial shift in global trade relations. The more recent trade agreements demonstrate a higher level of depth in terms of their coverage of trade policy areas, reflecting the growing complexity and scope of these agreements. As we observe rapid regional trade liberalization, most countries have become signatories to multiple trade agreements, leading to the formation of their own ego-centred networks of trade agreements, as highlighted by Manger et al. (2012) and Baldwin (2013). In a network of trade agreements, each country represents a node that is interconnected with other countries (nodes) through direct or indirect trade agreement relationships (Pauwelyn & Alschner, 2015). The dynamic growth of

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trade agreements networks has increasingly attracted attention to their economic impacts.

Since the primary objective of trade agreements is to facilitate market access among participating countries (Limão, 2016), existing studies have predominantly focused on investigating the impacts of trade agreements networks on trade flows. For example, Hur et al. (2010) conducted an analysis of trade agreement networks from 1960 to 2000 and found that countries positioned at the centre of these networks tend to export more than their counterparts. In a case study examining Israel, Deltas et al. (2012) demonstrated that the signing of a new trade agreement by an original member country, leading to the expansion of the network of trade agreements, increases bilateral trade between the original member countries. Furthermore, utilizing data from 96 countries over the 1960–2010 period, Sopranzetti (2018) found that the expansion of a trade agreements network does not necessarily enhance bilateral trade for all member countries. However, if a new trade agreement enhances the centrality of a member country's position in the network, it is likely to increase the country's bilateral trade with other member countries. Fan et al. (2024) examined the influence of trade agreement network centrality on bilateral participation in Global Value Chains (GVCs) among member countries using data from 43 countries spanning the years 2000–2014. They suggest that this impact depends on the relative magnitude of two opposing effects: the resource allocation effect and the market substitution effect.

In this paper, we aim to investigate how the centrality of a country within an ego-centred trade agreements network affects its level of economic complexity, focusing on the country's trade structure rather than its trade scale. A growing body of literature on product space theory views a country's economic development as a process of transforming economic complexity (Hausmann et al., 2007; Hidalgo & Hausmann, 2009; Hausmann & Hidalgo, 2011). This view is supported by empirical studies that have found a positive correlation between a higher level of economic complexity and better economic performance, measured in terms of per capita income and economic growth rate (Hausmann et al., 2007; Felipe et al., 2012; Hausmann et al., 2014; Poncet & de Waldemar, 2015; Braun, 2024). While fundamental factors like physical and human capital, labor, and natural resources offer limited explanations for a country's production and export patterns (Hausmann et al., 2007), more recent studies have considered the role of policy-related factors in determining economic complexity. These studies examine explanatory variables including foreign direct investment (Antonietti & Franco, 2021; Harding & Javorcik, 2012), cultural traits (Fan et al., 2018; Maurya & Sahu, 2022), preferential trade agreements (Stojčić et al., 2018), immigrant diversity (Bahar et al., 2022), institutional quality (Li, 2015; Vu, 2022), financial development (Nguyen et al., 2020), and the impacts of economic policy uncertainty (Hoang et al., 2023). While these investigations yield valuable insights into the policy-related determinants of economic complexity, little attention has been directed toward the role of trade agreements networks.¹

Motivated by existing research on both the effect of trade agreement networks on trade flows and the role of policy-related factors in shaping economic complexity, this paper aims to connect trade agreement networks with economic complexity. This topic has hardly been explored. To the best of our knowledge, Wang (2024) is the only existing study that examines the relationship between trade agreement networks and economic complexity. Wang's results indicate that the centrality of trade agreement networks enhances members' economic complexity by facilitating transnational knowledge diffusion. Given that the effectiveness of transnational knowledge diffusion varies with geographic and developmental characteristics of recipient and source countries (e.g., Bahar et al., 2014; Bitzer & Kerekes, 2008; Keller & Yeaple, 2013), our paper focuses on the conditioning effects of network member structure. Specifically, we examine the impact of a country's centrality within trade agreements network on its economic complexity, considering the geographic distance and developmental disparities between the country and its network partners.

The economic complexity of a country reflects both the diversification and ubiquity of its output, which are directly determined by the country's set of production capacities (Hausmann & Hidalgo, 2011; Bahar et al., 2022). Countries with a broader range of production capacities tend to exhibit a more diverse and less ubiquitous product portfolio, resulting in a higher level of economic complexity. Regarding the sources of production capabilities in an open economy, previous studies have highlighted the crucial role of transnational knowledge diffusion, facilitated by international interactions such as trade, foreign direct investment (FDI), and labor mobility (Blomström & Kokko, 1998; Bahar et al., 2014; Buckley et al., 2007; Orlic et al., 2018). Consequently, the centrality of a country's ego-centred trade agreement network is expected to influence the country's level of economic complexity, as it shapes the scale and direction of the flow of goods and production factors between the country and other member countries within the network.

Given that a country's centrality in its trade agreement network is primarily determined by both the number and depth of trade agreements (Sopranzetti, 2018; Fan et al., 2024; Laget, Osnago, Rocha, et al., 2020), we argue that an increase in a country's network centrality can enhance its economic complexity through two distinct mechanisms: breadth and depth effects. The breadth effect suggests that the country can benefit from the diffusion of knowledge across a broader spectrum of member countries due to the expansion of the trade agreement network. The depth effect implies that the country can acquire more profound knowledge diffusions from each individual member country due to the deepening of the trade agreement network. Both effects can expand the scope of a country's production capabilities through knowledge diffusion from the flow of goods and production factors between the country and its trade partners, thereby enhancing its economic complexity. However, since transnational knowledge diffusion is more likely to occur among countries with technological heterogeneity or gaps, an increase in a country's centrality functions more effectively when

¹ In this paper, we focus on trade agreement networks for several reasons. First, our primary goal is to investigate the role of policy-related factors in shaping economic complexity. Building a trade agreement network can be viewed as a trade policy or national strategy. For instance, China has proposed establishing a network of high-standard free trade zones, particularly with neighboring countries as part of the Belt and Road Initiative. Second, the trade agreement network not only shapes trade flows but also influences the movement of factors of production such as capital (through foreign direct investment) and labor mobility, which are crucial for transnational knowledge diffusion. Third, from an empirical research standpoint, trade agreement networks are more likely to be exogenous than trade networks in this study.

the country is geographically distant from its trade partners and when those partners are more developed.

By examining the relationship between the centrality of trade agreement networks and economic complexity, particularly focusing on the conditioning effects arising from the structure of network members, this paper contributes to the existing literature on the policy-related drivers of a country's economic complexity by highlighting the role played by the country's strategic positioning within its trade agreement network. Moreover, our research shifts the focus from the impact on trade scale to the impact on trade structure reflecting economic complexity, thus adding to previous studies on the effects of trade agreement networks. Additionally, our work also enriches the existing body of research on the determinants and impacts of transnational knowledge diffusion within the framework of trade agreement networks and economic complexity.

The remainder of this paper is organized as follows. The hypotheses concerning the impacts of centralizing trade agreement networks on economic complexity are developed in Section 2. The methodology and dataset used to test the hypotheses are explained in Section 3. The benchmark regression results and robustness checks, including accounting for potential endogeneity, are presented in Section 4. The main conclusions and policy implications are summarized in Section 5.

2. Theoretical analysis and hypotheses development

Hausmann and Hidalgo (2011) developed a model of economic complexity based on product space theory. They used a framework in which the production of each good requires a combination of complementary capabilities. More sophisticated goods require a larger set of capabilities. Therefore, the range of capabilities possessed by a country determines its production structure. The larger the range of capabilities, the more diverse the range of goods a country can produce, and the more goods produced by that country cannot be produced by others. In other words, the range of capabilities is positively associated with production diversification and negatively associated with production ubiquity. Economic complexity serves as a composite measure that captures the level of diversification and ubiquity of output. A more diversified and less ubiquitous production structure leads to a higher level of economic complexity. Thus, countries with more production capabilities tend to have a higher level of economic complexity. However, a critical question remains: how do countries acquire production capabilities?

A country's production capabilities can be developed through both internal and external channels. Transnational knowledge diffusion is widely recognized as a crucial external channel for acquiring and enhancing these capabilities. There are several mechanisms through which transnational knowledge diffusion occurs, facilitating the development of production capabilities. First, some knowledge is typically embodied in goods. By importing relevant goods, countries can learn the technologies and production methods through processes such as reverse engineering (Coe et al., 2009; Coe & Helpman, 1995; Rivera-Batiz & Romer, 1990). This enables importing countries to acquire new knowledge and improve their production capabilities. Second, foreign direct investment (FDI) is known to generate knowledge spillovers. When foreign firms invest in a host country, they bring in new technologies, management practices, and know-how, which can benefit local firms through horizontal and vertical linkages (Blomström & Kokko, 1998; Buckley et al., 2007; Orlic et al., 2018). These knowledge spillovers can contribute to the formation of local production capabilities. Finally, international labour mobility, particularly the movement of skilled workers, can directly transfer production capabilities to destination countries. Skilled workers who migrate to other countries bring their expertise, knowledge, and innovative ideas, which can significantly contribute to the development and enhancement of production capabilities in the receiving countries (Batista et al., 2007; Commander et al., 2004; Docquier; Rapoport, 2007; Moore et al., 2021; Zucker & Darby, 2007).

It is well-known that preferential trade agreements promote the flow of goods and production factors across member countries. Preferential trade agreements not only remove (or reduce) tariff and non-tariff barriers for goods and services, but also decrease bilateral trade policy uncertainty and the transaction costs associated with the trade of intermediates (Antràs & Staiger, 2012; Grossman, 2016; Limão, 2016). Moreover, deeper preferential trade agreements can yield more significant positive effects on the flow of goods and factors of production between participating countries (Ruta, 2018; Lee, 2019; Ornelas et al., 2021; Zhang et al., 2021). Given the interconnected nature of trade agreements within the network, they collectively shape the flow of goods and factors of production among member countries. This, in turn, can have a profound impact on the production capabilities of these countries through transnational knowledge diffusion.

In social network theory, a node's position within the network is crucial in accessing resources from other nodes. Similarly, the benefits of knowledge diffusion can vary depending on a country's position within a network of trade agreements (Hansen, 1999). The centrality of a country within a trade agreements network is typically used to identify its position. The centrality of a country within the network is mainly determined by two factors. First, the number of countries with which a country has signed trade agreements, which reflects the extent of direct connections the country has with other nodes in the network. Second, the depth of trade agreements signed by a country with other nodes in the network, which reflects the quality of the relationship between the country and other nodes. Therefore, a country that has deeper trade agreements with more countries is likely to occupy a more central position within the trade agreements network.

The centrality of a country's position within a trade agreements network has two principal effects on its production capabilities: the "breadth effect" and the "depth effect." The breadth effect arises from the diversity of trade partners associated with a higher centrality in the network. Countries with more central positions engage with a broader array of partners, thereby gaining exposure to a wide variety of goods and factors of production. Given that each country possesses unique capabilities (Hausmann & Hidalgo, 2011), this exposure facilitates the development of diverse production capabilities through transnational knowledge diffusion. The depth effect, on the other hand, pertains to the nature of the trade agreements themselves. Countries in central positions often participate in deeper trade agreements, which are more effective at fostering the movement of goods and production factors across member nations (Orefice & Rocha, 2014; Johnson & Noguera, 2017; Zeng et al., 2021). As a result, such countries are better positioned to enhance their

production capabilities through knowledge diffusion from their trade partners.

It is important to acknowledge, however, that the provisions of deep trade agreements can sometimes present short-term challenges to knowledge spillovers for certain member countries. For instance, the inclusion of stringent intellectual property (IP) protections in agreements between developed and developing countries may restrict the latter's access to technologies and innovations, disproportionately favoring developed members. Similarly, the harmonization of regulatory standards—such as product safety or environmental regulations—may limit a developing country's flexibility to adopt alternative standards or learn from diverse global practices.

Despite these challenges, deep trade agreements can create a long-term environment conducive to innovation and knowledge transfer in developing countries. By fostering a stable and predictable regulatory framework, such agreements encourage firms to invest in innovation and technological adoption (Campi & Dueñas, 2019; Hudson & Minea, 2013; Neves et al., 2021). Consequently, the centralization of a country's position within the trade agreements network can positively influence its economic complexity, as production capabilities are closely linked to economic complexity. Based on this discussion, we have a hypothesis as follows.

H1. *The centrality of a country within a network of trade agreements has a positive effect on its economic complexity.*

While the overall effect of the centrality of a country in a trade agreements network on its economic complexity is expected to be positive, the magnitude of this effect may vary depending on the structure of member countries within the network. Previous studies on knowledge diffusion have shown that significant parts of knowledge are disembodied or tacit, requiring more direct forms of human interaction for transnational diffusion to occur (Arrow, 1969; Polanyi, 1962). The farther the geographical distance between two countries, the more difficult it is for them to interact directly with each other, making knowledge diffusion more likely to take place between countries that are geographically closer to each other. In fact, transnational diffusion of knowledge decays with geographical distance (Bahar et al., 2014; Bottazzi & Peri, 2003; Jaffe et al., 1993; Keller, 2002; Keller & Yeaple, 2013). As a result, countries situated closer to one another geographically are more inclined to possess similar sets of production capabilities, whereas those farther apart are more likely to exhibit divergent production capabilities.

Inspired by this perspective and following Krugman's (1991) approach, we categorize trade agreements into two distinct groups: "natural" and "non-natural." The former refers to intra-regional trade agreements among geographically proximate countries, while the latter encompasses inter-regional trade agreements among countries with greater bilateral geographical separation. Using this classification, a country's trade agreements network can also be divided into "natural" and "non-natural" sub-networks. The former includes the country's "natural" trade agreement partners, while the latter comprises its "non-natural" trade agreement partners. Considering the similarity of production capabilities between member countries, we expect that, compared to the centrality of the "natural" trade agreements network, the centrality of the "non-natural" trade agreements network to have a greater marginal impact on economic complexity levels through both breadth and depth effects on the scope of production capabilities. Based on these ideas, we have the following hypothesis.

H2. *The centrality of a country within the "non-natural" network of trade agreements has a stronger marginal effect on its economic complexity compared to the centrality within a "natural" network.*

The configuration of a trade agreements network can also vary based on the economic development level of member countries, which may affect the impact of centrality within the network on economic complexity. Transnational knowledge diffusion is predicated on the presence of a technology gap between countries. A wider technology gap increases the likelihood of transnational knowledge transfer, as countries with less advanced technologies can adopt innovations from more developed nations (Findlay, 1978; Nelson & Phelps, 1966). Developed countries, characterized by advanced technological capabilities and innovation ecosystems, often serve as primary sources of knowledge diffusion. In contrast, developing countries tend to exhibit lower technological prowess and innovation capacities, thus limiting their ability to generate substantial technological spillover (Blomström & Sjöholm, 1999; Bitzer & Kerekes, 2008; Coe et al., 2009). However, by strategically forming trade agreements with technologically advanced nations, developing countries can enhance their access to new technologies and accelerate their economic complexity.

Building upon this understanding, we separate a country's trade agreements network into two distinct sub-networks: "developed" and "developing." The former refers to the sub-network comprising the country's relatively more developed trade agreement partners, and the latter refers to the sub-network comprising the country's relatively less developed trade agreement partners. Given that developed member countries are more likely to generate transnational knowledge diffusion, we anticipate that, in comparison to the centrality of the "developing" trade agreements network, the centrality of the "developed" trade agreements network will exert a more pronounced marginal impact on the economic complexity level through its breadth and depth effects on the range of production capabilities. Based on the above discussion, we have a hypothesis as follows.

H3. *The centrality of a country within the network of "developed" trade agreements has a stronger marginal effect on its economic complexity compared to the centrality within a network of "developing" trade agreements.*

3. Main variables and data

3.1. Main variables and measurements

We calculated a country's economic complexity index (ECI) using the reflection method proposed by Hausmann and Hidalgo (2011). Let $k_{i,0}$ denote the diversification of products exported by country i , and $k_{p,0}$ represent the ubiquity of product p . $k_{i,0}$ and $k_{p,0}$ are

given by

$$k_{i,0} = \sum_p M_{i,p} \quad (1)$$

$$k_{p,0} = \sum_i M_{i,p} \quad (2)$$

where $M_{i,p}$ denotes whether country i exports product p . $M_{i,p} = 1$ if country i exports the good p ; 0 otherwise. A higher value of $k_{i,0}$ represents higher level of export diversity for country i . Similarly, a higher value of $k_{p,0}$ represents higher level of ubiquity of product p .

By accounting for the ubiquity of each product in the level of export diversification of country i , we can calculate the ubiquity weighted export diversification of country i as follows:

$$k_{i,n} = \frac{1}{k_{i,0}} \sum_p M_{i,p} k_{p,n-1} \quad (3)$$

where, $k_{p,n-1}$ is given by

$$k_{p,n-1} = \frac{1}{k_{p,0}} \sum_i M_{i,p} k_{i,n-2} \quad (4)$$

Substituting equation (4) into equation (3), and repeating the iterative process until $k_{i,n} = k_{i,n-2}$, yields

$$k_{i,n} = \sum_i \tilde{M}_{i,i} k_{i,n-2} \quad (5)$$

where $\tilde{M}_{i,i} = \frac{M_{i,p} M_{i,p}}{\sum_p k_{i,0} k_{p,0}}$.

Using $\vec{k}$ to represent the second largest eigenvector of $\tilde{M}_{i,i}$,² and using $\langle \vec{k} \rangle$ and $stdev(\vec{k})$ to denote the $\vec{k}$'s mean and standard deviation, respectively, we obtain the economic complexity of country i as follows.

$$ECI_i = \frac{\vec{k} - \langle \vec{k} \rangle}{stdev(\vec{k})} \quad (6)$$

Based on social network theory, the centrality of a country in its network of trade agreements is measured using eigenvector centrality, also known as hub centrality. This measure captures the importance of the country within the network by considering the extent of its connections and the influence exerted by its trading partners. Specifically, it is given by

$$Centrality_i = \lambda^{-1} \sum_j Depth_{ij} e_j \quad (7)$$

where $Centrality_i$ is the centrality of country i in its trade agreements network. $Depth_{ij}$ is the depth of the trade agreement between countries i and j , which is measured using the method proposed by Hofmann et al. (2017).³ e_j is the eigenvector centrality of country j in its trade agreements network. λ is the largest eigenvalue of the eigenvector incorporating all nodes.

As shown in equation (7), a country's centrality level in its trade agreements network is influenced by three crucial factors: (i) the number of countries with which the country has established trade agreements, reflecting the extent of its direct connections with other nodes in the network; (ii) the depth of trade agreements signed by the country with other countries in the network, representing the quality of the relationship between the country and other nodes; (iii) the centrality of the country's trade agreement partners in their respective networks, highlighting the significance of the country's trade agreement partners. According to the social network theory, all three factors are all positively associated with a country's centrality in its trade agreements network.

In line with prior research on the determinants of economic complexity (e.g., Bahar et al., 2022; Zhou et al., 2023), we incorporate several control variables as follows: (i) population size (*POP*), measured in ten million people per country. As a larger population is known to facilitate technological progress (Kremer, 1993), we expect the estimated coefficient of *POP* to be positive; (ii) Human capital (*HC*), assessed by the enrolment rate of colleges. Since human capital accumulation is crucial for fostering innovation (Dustmann et al., 2011), we anticipate a positive coefficient for *HC*; (iii) Natural resource rent (*NatR*), calculated as the percentage of GDP derived from rents of oil, natural gas, coal, forest, and mineral resources. Given that economies abundant in natural resources may have less incentive to pursue innovation (Hoang et al., 2023), this factor is expected to have a dampening effect on economic growth; (iv) Financial development (*FIN*), measured by the ratio of domestic credit to private sector to gross domestic product (GDP). Given that

² See Hidalgo and Hausmann (2009).

³ Hofmann et al. (2017) employed a methodology whereby 48 provisions of trade agreements were categorized as either present (coded as 1) or absent (coded as 0) within a given trade agreement. The depth of a particular trade agreement was subsequently determined by summing the number of provisions that were included within the agreement.

external financing supports high-tech industries (Nguyen et al., 2020), we expect this variable to positively influence economic complexity. In addition, to account for unobserved heterogeneity, we also include the time and country fixed effects in our regression equations.

3.2. Data and descriptive statistics

The data used in this study are sourced from several databases as shown in the Appendix. Economic complexity data is obtained from the Atlas database of Harvard University, which utilizes the four-digit product code of the SITC classification standard. The depth of trade agreements data is collected from the World Bank database (World Bank, 2023). The centrality of the trade agreements network is calculated using Pajek software (Pajek, 2022). The range of production capabilities is computed using data on labor employment categories gathered from the US Bureau of Labor Statistics (BLS, 2022). Although the data is originally based on the NAICS industry standard, it is converted to the SITC classification standard to align with the measurement of economic complexity. Additionally, data on population size, openness, college enrollment rate, natural resource rent, and financial development were sourced from the World Development Indicators Database (WDI, 2022). After ensuring consistency across all variables, the resulting dataset includes 78 countries and covers the period from 2000 to 2019. To mitigate the potential impact of outliers in the sample, a two-tailed Winsorizing approach was applied to all continuous variable data at the 1% level.

The descriptive statistics for the variables are presented in Table 1. Certain variables such as Economic Complexity Index (ECI), Centrality of trade agreement network (Centrality), and Population Size (POP), exhibit a significant deviation between their means and medians, large standard deviations, and notable differences between maximum and minimum values. These characteristics indicate a higher level of dispersion in the data for these variables. Moreover, the Pearson correlation coefficients between most explanatory variables are very low. The strongest correlation observed is between human capital and financial development, with a correlation coefficient of 0.669. Additionally, the variance inflation factor (VIF) for all variables is less than 10, which implies the absence of statistically significant multicollinearity in the regression model. Finally, we employ the Fisher-PP (Phillips-Perron) test to check for the presence of a unit root in the panel data. The Fisher-PP test results indicate the rejection of the null hypothesis, suggesting that the panel data are stationary.

Table 1
Descriptive statistics and correlation analysis of variables.

	ECI	Centrality	POP	HC	NatR	FIN
<i>ECI</i>	1	0.614***	0.061**	0.737***	−0.622***	0.711***
<i>Centrality</i>		1	−0.022	0.533***	−0.584***	0.593***
<i>POP</i>			1	−0.033	0.130***	−0.012
<i>HC</i>				1	−0.501***	0.669***
<i>NatR</i>					1	−0.504***
<i>FIN</i>						1
Mean	0.121	0.056	3.438	0.405	0.065	0.595
Median	0.005	0.006	1.286	0.379	0.025	0.474
S.D.	0.945	0.069	5.317	0.274	0.097	0.456
Minimum	−1.694	0	0.0615	0.014	0.0003	0.042
25% quartile	−0.592	0.0006	0.553	0.165	0.005	0.227
75% quartile	0.851	0.1199	4.033	0.621	0.082	0.877
Maximum	2.168	0.219	31.605	1.092	0.463	1.845
VIF		1.75	1.28	1.81	1.24	1.82
Fisher-PP	0.000	0.000	0.043	0.000	0.000	0.000

Note: Correlation analysis uses the Pearson correlation coefficients. The unit of the variable *POP* is ten million. Capability values are in natural logarithms. *ECI* is economic complexity index, *Centrality* is the centrality of trade agreement network, *POP* is population size, *HC* is human capital level, *NatR* is natural resource rent, and *FIN* is financial development level. ** and *** represent significance at the 5% and 10% level, respectively.

4. Empirical results and discussion

4.1. The overall effect of centralizing trade agreement networks

In this section, we first testing hypothesis H1, which suggests that centralization of a country's position in its trade agreements network has a positive effect on its economic complexity level. Building upon the variables introduced above, we construct a regression equation as follows.

$$ECI_{it} = \beta_0 + \beta_1 \text{Centrality}_{it} + \beta_2 X + \mu_i + \vartheta_t + \varepsilon_{it} \quad (8)$$

where i and t represent the country and year, respectively. ECI_{it} is the economic complexity level of country i in year t , and Centrality_{it} is the centrality of country i in its trade agreements network in year t . X is the vector of control variables, which includes population size, openness degree, human capital, natural resource rent, and financial development level. μ_i is the country fixed effect, which are used to control time-invariant heterogeneity across exporting countries. ϑ_t is the time fixed effect, which captures other time-variant factors. ε_{it} is the usual error term.

Equation (8) estimation results are presented in Table 2, where column 1 shows the results without including the control variables. The estimated coefficient for the centrality of the trade agreements network is positive and statistically significant at the 10% level. When the control variables are included in the regression equation, as shown in column 2 of Table 2, the estimated coefficient for the centrality of the trade agreements network remains positive and becomes statistically significant at the 1% level. Specifically, a one-unit increase in a country's trade agreements network centrality is associated with a 0.772-unit increase in economic complexity, thereby supporting hypothesis H1.

These results imply that countries with higher centrality within their trade agreement networks tend to exhibit greater economic complexity, which is also in line with Wang's (2024) work. This finding can be attributed to the positive correlation between a country's production capabilities and its economic complexity. A centralized position in the trade agreements network facilitates transnational knowledge diffusion by establishing more extensive and deeper linkages with other member countries, thus expanding the range of the country's production capabilities and promoting its economic complexity, as evidenced by the "breadth effect" and "depth effect."

Regarding the control variables, the estimated coefficient for population size is significantly positive at the 1% level, which aligns with expectations. This result is also consistent with Hoang et al. (2023). However, the estimated coefficients for other variables are not statistically significant. In column 2, an adjusted R-squared value of 0.965 indicates that the model provides a very good fit to the data, with the independent variables collectively accounting for a large portion of the variability in economic complexity.

4.2. The conditioning effects of network structure

This section focuses on testing hypotheses H2 and H3, which examine the conditioning impacts of the centrality of a country within the trade agreements network on its economic complexity. Specifically, we investigate whether the centrality of "non-natural" and "developed" trade agreements networks has a more pronounced effect on economic complexity compared to the centrality of "natural" and "developing" trade agreements networks, respectively.

To identify a country's "natural" and "non-natural" trade agreements networks, we consider the geographic proximity between the country and its trade partners. If the geographic distance between a country and its trade partners is greater (or less) than the average level, we classify these trade partners as belonging to the country's "non-natural" (or "natural") trade agreements network. Moreover, to distinguish between a country's "developed" and "developing" trade agreements networks, we utilize the income-level classifications established by the World Bank (WB).⁴ Specifically, if a country's trade partners are classified as high-income (or low-income) economies according to the WB's standards, they are grouped into the country's "developed" (or "developing") trade agreements network. To ensure alignment with our theoretical framework, we excluded oil-abundant countries from the high-income category.

Based on Equation (7), which defines the centrality of the trade agreement network, we create two pairs of centrality variables: $N_Centrality$ and $NN_Centrality$ to represent the centrality of the "natural" and "non-natural" trade agreement network, respectively, and $ED_Centrality$ and $ING_Centrality$ to represent the centrality of the "developed" and "developing" trade agreement network, respectively. Subsequently, using Equation (9), we formulate two regression models to test hypotheses H2 and H3 as follows:

$$ECI_{it} = \beta_0 + \beta_1 N_Centrality_{it} + \beta_2 NN_Centrality_{it} + \beta_3 X + \mu_i + \vartheta_t + \varepsilon_{it} \quad (9)$$

$$ECI_{it} = \beta_0 + \beta_1 ED_Centrality_{it} + \beta_2 ING_Centrality_{it} + \beta_3 X + \mu_i + \vartheta_t + \varepsilon_{it} \quad (10)$$

⁴ According to the WB classification, our sample includes 37 high-income economies, excluding those classified as oil-abundant such as Saudi Arabia (SAU), Kuwait (KWT), Qatar (QAT), Oman (OMN), and Bahrain (BHR). These economies are Australia (AUS), Austria (AUT), Belgium (BEL), Bulgaria (BGR), Bahrain (BHR), Canada (CAN), Switzerland (CHE), Chile (CHL), Cyprus (CYP), Czech Republic (CZE), Germany (DEU), Denmark (DNK), Spain (ESP), Estonia (EST), Finland (FIN), France (FRA), United Kingdom (GBR), Greece (GRC), Croatia (HRV), Hungary (HUN), Ireland (IRL), Israel (ISR), Italy (ITA), Japan (JPN), South Korea (KOR), Kuwait (KWT), Lithuania (LTU), Latvia (LVA), Netherlands (NLD), Norway (NOR), New Zealand (NZL), Oman (OMN), Panama (PAN), Poland (POL), Portugal (PRT), Qatar (QAT), Romania (ROU), Saudi Arabia (SAU), Slovakia (SVK), Slovenia (SVN), Sweden (SWE), United States (USA).

Table 2
The overall effect of centralizing trade agreement networks.

	(1)	(2)
<i>Centrality</i>	0.406*	0.772***
	(0.212)	(0.223)
<i>POP</i>		0.091***
		(0.022)
<i>HC</i>		0.010
		(0.113)
<i>NatR</i>		−0.374
		(0.276)
<i>FIN</i>		0.018
		(0.046)
<i>Constant</i>	0.099***	−0.188*
	(0.013)	(0.105)
<i>Country FE</i>	YES	YES
<i>Year FE</i>	YES	YES
<i>Adj R-squared</i>	0.961	0.965
<i>Observations</i>	1483	1289

Note. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Values in parentheses underneath the estimated coefficients are the corresponding robust standard errors. *Centrality* is the centrality of trade agreement network, *POP* is population size, *HC* is human capital level, *NatR* is natural resource rent, and *FIN* is financial development level.

Equations (9) and (10) estimation results are presented in Table 3, where the estimated effects of *N_Centrality* and *NN_Centrality* are shown in columns 1 and 2, respectively. In column 1, where the control variables are excluded, the estimated coefficients for both *N_Centrality* and *NN_Centrality* are significantly positive. However, as shown in column 2, when control variables are introduced into the regression, only the estimated coefficient for *NN_Centrality* remains statistically significant at the 1% level. These findings support hypothesis H2, suggesting that because neighboring countries often have similar production capabilities, an increase in the centrality of a country within the “natural” trade agreements network formed by its neighboring countries results in limited transnational knowledge diffusion from both broad and deep perspectives. In contrast, non-neighboring countries often have different production capabilities, so an increase in the centrality of a country within the “non-natural” trade agreements network formed by its non-neighboring countries results in significant transnational knowledge diffusion from both broad and deep perspectives. Therefore, the centrality of a country within the “non-natural” trade agreements network has a more pronounced impact on the country’s economic complexity level compared to the centrality of a country within the “natural” trade agreements network.

Columns 3 and 4 of Table 3 present the estimated effects of *ED_Centrality* and *ING_Centrality*, respectively. In both cases, the estimated coefficients for *ED_Centrality* are significantly positive at the 1% level, while those for *ING_Centrality* are statistically insignificant. These results lend support to hypothesis H3, indicating that because developed countries often serve as primary sources of transnational knowledge diffusion, an increase in the centrality of a country within the “developed” trade agreements network formed by its developed partners can help the country obtain significant transnational knowledge diffusion from both broad and deep perspectives. By contrast, an increase in the centrality of a country within the “developing” trade agreements network formed by its developing partners results in limited transnational knowledge diffusion from both broad and deep perspectives. Therefore, compared to the centrality of a country within the “developing” trade agreements network, the centrality of a country within the “developed” trade agreements network has a more pronounced impact on the country’s economic complexity level.

4.3. Robustness testing

To test the robustness of the benchmark regression results presented in Tables 2 and 3, we employ four estimation strategies.

First, we assess whether the estimation results are affected by the product classification standard used to calculate the level of economic complexity. Specifically, we recalculate the economic complexity level by replacing the SITC classification standard with the HS92 classification standard. The data for this exercise are sourced from the Atlas database of Harvard University. The robustness test results for hypothesis H1 are presented in column 1 of Table 4, and for hypotheses H2 and H3, they are shown in column 1 of Tables 5 and 6, respectively. As indicated, the estimation results remain consistent with those presented in Tables 2 and 3, suggesting that our benchmark findings are not sensitive to the economic complexity measurement method.

Second, to address potential estimation bias caused by missing variables in our benchmark regressions, we control for country and time fixed effects. To further mitigate the possibility of missing variable bias, we include an additional control variable: bilateral investment treaties (BITs). BITs have been shown to significantly impact bilateral trade and direct investments among member countries (Boffa et al., 2019), which may affect economic complexity through the facilitation of transnational knowledge diffusion. The robustness testing results for hypothesis H1 are presented in column 2 of Table 4, while the estimation results for hypotheses H2 and H3 are presented in column 2 of Tables 5 and 6, respectively. As shown, the estimation results are consistent with the estimation results presented in Tables 2 and 3. This robustness check suggests the absence of significant missing variable bias in our previous benchmark regressions.

Table 3
The conditioning effects of network structure (testing hypothesis H2 and H3).

	(1)	(2)	(3)	(4)
<i>N_Centrality</i>	0.279*** (0.106)	0.089 (0.101)		
<i>NN_Centrality</i>	0.329*** (0.076)	0.298*** (0.072)		
<i>ED_Centrality</i>			2.487*** (0.727)	3.514*** (1.292)
<i>ING_Centrality</i>			−0.540* (0.307)	−0.202 (0.370)
<i>POP</i>		0.162*** (0.027)		0.021** (0.009)
<i>HC</i>		0.294** (0.124)		0.029 (0.096)
<i>NatR</i>		−0.058 (0.293)		0.490 (0.569)
<i>FIN</i>		0.060 (0.049)		−0.052 (0.053)
<i>Constant</i>	0.195*** (0.007)	−0.472*** (0.127)	0.458*** (0.068)	0.252* (0.143)
<i>Country FE</i>	YES	YES	YES	YES
<i>Year FE</i>	YES	YES	YES	YES
<i>Adj R-squared</i>	0.960	0.966	0.974	0.978
<i>Observations</i>	1262	1084	817	711

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Values in parentheses underneath the estimated coefficients are the corresponding robust standard errors. *N_Centrality* is centrality of natural trade agreement network, *NN_Centrality* is centrality of non-natural trade agreement network, *ED_Centrality* is centrality of developed country trade agreement network, *ING_Centrality* is centrality of developing country trade agreement network, *POP* is population size, *HC* is human capital level, *NatR* is natural resource rent, and *FIN* is financial development level.

Table 4
Robustness testing results for hypothesis H3.

	(1)	(2)	(3)	(4)
<i>Centrality</i>	0.656*** (0.210)	0.498** (0.235)	0.820*** (0.230)	0.410* (0.243)
<i>POP</i>	0.088*** (0.023)	0.130*** (0.022)	0.092*** (0.022)	0.086*** (0.023)
<i>HC</i>	0.038 (0.104)	0.212* (0.119)	−0.071 (0.111)	0.022 (0.137)
<i>NRR</i>	−0.246 (0.256)	−0.351 (0.278)	−0.406 (0.277)	−0.347 (0.285)
<i>FIN</i>	0.038 (0.051)	0.036 (0.052)	0.008 (0.050)	0.065 (0.069)
<i>BIT</i>		−0.008*** (0.002)		
<i>Constant</i>	−0.203* (0.109)	−0.129 (0.122)	−0.153 (0.108)	−0.348*** (0.117)
<i>Country FE</i>	YES	YES	YES	YES
<i>Year FE</i>	YES	YES	YES	YES
<i>Adj R-squared</i>	0.970	0.966	0.966	0.956
<i>Observations</i>	1289	1151	1175	1100

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Values in parentheses underneath the estimated coefficients are the corresponding robust standard errors. *N_Centrality* is centrality of natural trade agreement network, *NN_Centrality* is centrality of non-natural trade agreement network, *ED_Centrality* is centrality of developed country trade agreement network, *ING_Centrality* is centrality of developing country trade agreement network, *POP* is population size, *HC* is human capital level, *NatR* is natural resource rent, *FIN* is financial development level, and *BITs* are bilateral investment treaties.

Third, our sample period includes the 2008 global financial crisis (GFC) and the subsequent international trade collapse. These events could potentially distort the underlying effects of trade agreements on economic complexity, as the dynamics during and after the financial crisis might reflect short-term adaptations and crisis management strategies rather than the usual strategic economic interactions dictated by trade agreements. To ascertain the influence of these events, we follow the methodology of Lee (2019) and re-estimate after excluding data for 2008 and 2009. The robustness testing results for hypothesis H1 are presented in column 3 of Table 4, while the results for hypotheses H2 and H3 are presented in column 3 of Tables 5 and 6, respectively. As shown, the estimation results are consistent with the results presented in Tables 2 and 3. This indicates that our findings are robust and not significantly impacted by the GFC.

Table 5
Robustness testing results for hypothesis H2.

	(1)	(2)	(3)	(4)
<i>N_Centrality</i>	−0.036 (0.087)	0.007 (0.120)	0.071 (0.100)	−0.060 (0.140)
<i>NN_Centrality</i>	0.373*** (0.076)	0.307*** (0.075)	0.295*** (0.074)	0.512*** (0.119)
<i>POP</i>	0.165*** (0.025)	0.162*** (0.028)	0.165*** (0.028)	0.167*** (0.027)
<i>HC</i>	0.478*** (0.110)	0.353*** (0.132)	0.221* (0.121)	0.300* (0.157)
<i>NatR</i>	0.059 (0.279)	−0.204 (0.303)	−0.098 (0.291)	−0.036 (0.304)
<i>FIN</i>	0.077 (0.054)	0.060 (0.054)	0.048 (0.054)	0.177*** (0.068)
<i>BIT</i>		−0.009*** (0.002)		
<i>Constant</i>	−0.036 (0.087)	0.007 (0.120)	0.071 (0.100)	−0.060 (0.140)
<i>Country FE</i>	YES	YES	YES	YES
<i>Year FE</i>	YES	YES	YES	YES
<i>Adj R-squared</i>	0.972	0.966	0.968	0.958
<i>Observations</i>	1084	975	993	915

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Values in parentheses underneath the estimated coefficients are the corresponding robust standard errors. *N_Centrality* is centrality of natural trade agreement network, *NN_Centrality* is centrality of non-natural trade agreement network, *ED_Centrality* is centrality of developed country trade agreement network, *ING_Centrality* is centrality of developing country trade agreement network, *POP* is population size, *HC* is human capital level, *NatR* is natural resource rent, *FIN* is financial development level, and *BITs* are bilateral investment treaties.

Table 6
Robustness testing results for hypothesis H3.

	(1)	(2)	(3)	(4)
<i>ED_Centrality</i>	3.078** (1.230)	3.559*** (1.312)	3.526*** (1.322)	2.808* (1.440)
<i>ING_Centrality</i>	−0.015 (0.322)	−0.207 (0.362)	−0.331 (0.393)	−0.024 (0.458)
<i>POP</i>	0.021** (0.009)	0.020** (0.009)	0.022** (0.010)	0.019** (0.009)
<i>HC</i>	−0.028 (0.089)	0.013 (0.097)	0.052 (0.111)	0.015 (0.120)
<i>NatR</i>	1.447* (0.738)	0.517 (0.573)	0.242 (0.632)	0.428 (0.585)
<i>FIN</i>	−0.051 (0.052)	−0.044 (0.056)	−0.074 (0.058)	−0.050 (0.072)
<i>BIT</i>		0.002 (0.002)		
<i>Constant</i>	0.235* (0.140)	0.164 (0.156)	0.262* (0.149)	0.254** (0.121)
<i>Country FE</i>	YES	YES	YES	YES
<i>Year FE</i>	YES	YES	YES	YES
<i>Adj R-squared</i>	0.981	0.977	0.978	0.976
<i>Observations</i>	711	696	615	582

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Values in parentheses underneath the estimated coefficients are the corresponding robust standard errors. *N_Centrality* is centrality of natural trade agreement network, *NN_Centrality* is centrality of non-natural trade agreement network, *ED_Centrality* is centrality of developed country trade agreement network, *ING_Centrality* is centrality of developing country trade agreement network, *POP* is population size, *HC* is human capital level, *NatR* is natural resource rent, *FIN* is financial development level, and *BITs* are bilateral investment treaties.

Finally, our empirical analysis includes European Union (EU) member countries. The EU represents a unique economic and political union of 27 countries characterized by profound integration in terms of the depth and breadth of economic ties. This integration has a significant influence on trade flows and other economic dimensions among member countries. To examine the impact of including EU member countries on our estimation results, we exclude data from the 12 countries that constituted the European

Community, the predecessor of the EU.⁵ The robustness testing results for hypothesis H1 are presented in column 4 of Table 4, and for hypotheses H2 and H3, they are shown in column 4 of Tables 5 and 6, respectively. As shown, the estimation results are consistent with the results presented in Tables 2 and 3, affirming that the inclusion of EU members does not bias our benchmark regression results.

4.4. Accounting for potential endogeneity

The empirical analysis presented thus far has been conducted under the assumption that the centrality of a country's trade agreements network is exogenous. However, it is crucial to take into account the potential existence of reverse causality between the centrality of the trade agreements network and economic complexity. In particular, as a country's economic complexity increases, its products may become more competitive in international markets. This enhanced competitiveness may necessitate the deepening of existing trade agreements or the establishment of new deep agreements with additional countries, which, in turn, can result in a more central position for the country within its trade agreements network. To address the potential issue of endogeneity arising from reverse causality, we employ an instrumental variable approach.

Building on previous research, including Alesina et al. (2016) and Bahar et al. (2022), we construct an instrument by predicting the depth of trade agreements between member countries. To minimize the potential violation of the exclusion restriction, we use a parsimonious gravity model to predict the depth of trade agreements. This gravity model takes into account common cultural and historical characteristics among the member countries. The formulation of the gravity model is as follows:

$$Depth_{ijt} = \delta_0 + \delta_1 Dist_{ij} * 2007_t + \delta_2 Lang_{ij} * 2007_t + \delta_3 Colony_{ij} * 2007_t + \delta_4 Border_{ij} * 2007_t + \omega_{ijt} + \varepsilon_{ijt} \quad (11)$$

where $Dist_{ij}$ is the bilateral weighted geographic distance between countries i and j . $Lang_{ij}$, $Colony_{ij}$ and $Border_{ij}$ are dummy variables indicating the presence of a common official language, previous colonial ties, and common border between countries i and j , respectively. To account for significant growth in global trade agreements in 2007 in our sample, we constructed a dummy variable, 2007_t , and included its interaction with all variables in equation (11). If the period t is after the year 2007, 2007_t takes the value of 1; 0 otherwise. ω_{ijt} is a dummy variable capturing time-variant heterogeneity across country pairs. Finally, ε_{ijt} is the usual error term.

Using the predicted depth of trade agreements derived from estimating equation (11), we compute the predicted centrality of the trade agreements network based on equation (7). The centrality of trade agreements networks is expected to correlate with these predicted values. Moreover, because the estimated centrality is less influenced by the economic factors of exporting countries, it is likely exogenous and satisfies the exclusion restriction requirement. Accordingly, we use the predicted centrality as an instrumental variable, denoted as $IV_Centrality1$, for the centrality of the trade agreements network.

Applying the two-stage least squares (2SLS) methodology, we re-estimate the impact of trade agreements network centrality on economic complexity. The results, presented in Column 1 of Table 7, show that the coefficient for trade agreements network centrality remains positive and statistically significant at the 1% level. These findings align with the benchmark regression results reported in Table 2, reinforcing the robustness of the observed relationship between trade agreements network centrality and economic complexity.

To further address endogeneity concerns in the benchmark estimation, we adopt the approach proposed by Nunn and Qian (2014) to construct an alternative instrumental variable, $IV_Centrality2$. This variable is defined as the product of a country's average geographic distance to other countries and the average centrality of other countries in the trade agreements network. Regarding relevance, countries geographically closer to highly central trade partners are more likely to form trade agreements with them (Anderson & Van Wincoop, 2003; Baier & Bergstrand, 2004; Borchert et al., 2022). Such agreements facilitate deeper integration into the trade network, enhancing the country's own centrality within the network. In terms of exogeneity, geographic distance is a fixed, natural characteristic unaffected by economic conditions or policies, while the average centrality of other countries in the network is external to any specific country's domestic circumstances. Estimation results using $IV_Centrality2$, presented in Column 2 of Table 7, confirm that the coefficient for trade agreements network centrality remains positive and statistically significant at the 1% level.

Additionally, we apply the methodology developed by Lewbel (2012) to construct an alternative instrumental variable for trade agreements network centrality. This approach leverages the heteroskedastic structure of the error term to generate instruments, enabling the identification of structural parameters in models with endogenous regressors. Using this method, we construct $IV_Centrality3$ by employing four control variables: population size, human capital level, natural resource rents, and financial development level. The corresponding 2SLS estimation results, presented in Column 3 of Table 7, indicate that the coefficient for trade agreements network centrality remains positive and statistically significant at the 1% level.

To validate these instrumental variable estimations, we conduct the Kleibergen-P LM-test and the Kleibergen-P Wald F-test. These tests confirm the absence of underidentification and weak instrument issues across all three specifications. Collectively, the empirical findings from these instrumental variable analyses demonstrate that our benchmark regression results are robust to potential endogeneity concerns, including reverse causality between a country's centrality in the trade agreements network and its economic complexity.

⁵ The 12 member countries of the European Community include Belgium, Denmark, France, Germany, Greece, Ireland, Italy, Luxembourg, Netherlands, Portugal, Spain, the United Kingdom.

Table 7
2SLS estimation results using an instrumental variable.

Variables	(1)	(2)	(3)
	IV_Centrality1	IV_Centrality2	IV_Centrality3
<i>Centrality</i>	1.137*** (0.316)	1.010* (0.583)	1.641*** (0.341)
<i>POP</i>	0.091*** (0.022)	0.091*** (0.022)	0.090*** (0.022)
<i>HC</i>	0.003 (0.114)	0.006 (0.112)	−0.007 (0.115)
<i>NatR</i>	−0.390 (0.275)	−0.374 (0.281)	−0.413 (0.280)
<i>FIN</i>	0.019 (0.045)	0.019 (0.045)	0.022 (0.045)
<i>Country FE</i>	YES	YES	YES
<i>Year FE</i>	YES	YES	YES
<i>Observations</i>	1289	1273	1289
<i>K-P rk LM</i>	184.164 [0.000]	61.343 [0.000]	151.176 [0.000]
<i>K-P rk Wald F</i>	424.215 {16.38}	80.460 {16.38}	199.786 {10.27}

Note. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Values in parentheses underneath the estimated coefficients are the corresponding robust standard errors. Values in square parenthesis, [], are also the estimated *p*-values. Values in {} are Stock-Yogo weak-identification test critical values. *Centrality* is the centrality of trade agreement network, *POP* is population size, *OPEN* is openness degree, *HC* is human capital level, *NatR* is natural resource rent, and *FIN* is financial development level. *IV_Centrality1*, *IV_Centrality2*, and *IV_Centrality3* represent the instrumental variables derived from the three respective methods.

5. Conclusion and policy implications

This paper investigates the impact of a country's centrality within the trade agreements network on its economic complexity, emphasizing the conditioning effects of network structure. We argue that a more centralized position within the trade agreements network can enhance a country's production capabilities by facilitating transnational knowledge spillovers. This centralization promotes both a "breadth effect," where knowledge diffusion occurs across a broad spectrum of member countries, and a "depth effect," where deeper knowledge spillovers are acquired from individual member countries. As a country's production capabilities are positively associated with its economic complexity, a positive relationship between the country's centrality within the trade agreements network and its economic complexity is predicted.

However, this relationship can be influenced by the structure of the trade agreements network, a phenomenon we refer to as the conditioning effect of network structure. Specifically, we propose that the centrality of "non-natural" trade agreements networks, which consist of interregional member countries, has a more pronounced impact on economic complexity compared to "natural" trade agreements networks composed of intraregional member countries. The rationale is that transnational knowledge diffusion diminishes with geographical proximity, leading neighboring countries to have similar production capabilities. Consequently, the centrality of "natural" trade agreements networks has a weaker impact on knowledge diffusion from both the breadth and depth perspectives. Additionally, we contend that the centrality of "developed" trade agreements networks, comprising developed member countries, exerts a stronger effect on economic complexity compared to "developing" trade agreements networks, given that knowledge diffusion is more likely to originate from developed countries.

To test these hypotheses, we analyze data from 78 countries over the 2000–2019 period, sourced from the Atlas of Economic Complexity database (Harvard University) and the World Development Indicators database (World Bank). The centrality of each country in the trade agreements network is calculated using Pajek software. The empirical analysis confirms our predictions, and the robustness of the results is verified through multiple tests, including checks for endogeneity.

The findings presented in this paper have significant policy implications for developing countries aiming to enhance their economic complexity through trade agreements networks. First, developing countries should strive to increase both the number and depth of their trade agreements to enhance their centrality within the network and facilitate structural transformation. Second, it is crucial for developing countries to prioritize forming "non-natural" trade agreements with interregional partners rather than relying heavily on neighboring countries. Such networks are more effective in generating diverse and impactful knowledge spillovers. Third, including developed countries in trade agreements networks is vital, as these countries serve as primary sources of advanced knowledge and technology. Engaging with developed countries can significantly accelerate the economic complexity of developing nations. By strategically building and deepening trade agreements networks, developing countries can better leverage transnational knowledge diffusion to achieve economic transformation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhaobin Fan reports financial support was provided by National Social Sciences fund of China.

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Appendix

Appendix

Definitions and data sources for main variables

Variables	Variables definition and measurement	Data source
Centrality of trade agreements (Centrality)	Centrality of a country in its trade agreement networks, measured using eigenvector centrality	World bank Database
Economic complexity index (ECI)	A composite measure that captures the level of diversification and ubiquity of outputs.	Atlas database of Harvard University
Production capabilities (Capability)	Range of production capabilities, measured by the average number of labour employment categories involved in producing a good	The US Bureau of Labor and Statistics
Population size (POP)	Total population	WDI Database
Human capital (HC)	Enrolment rate of colleges	WDI Database
Openness degree (OPEN)	Ratio of total trade of a country to its GDP	WDI Database
Natural resource rent (NatR)	The percentage of GDP derived from rents of oil, natural gas, coal, forest, and mineral resources.	WDI Database
Financial development (FIN)	The ratio of domestic credit to private sector to GDP	WDI Database
Bilateral investment agreements (BITs)	Number of bilateral investment agreements	IIAN Database
Geographic distance (Dist)	Bilateral weighted geographic distance between countries	CEPII Database
Common language (Lang)	Presence of a common official language	CEPII Database
Colonial relationship (Colony)	Previous colonial ties	CEPII Database
Common Border (Border)	Common border between countries	CEPII Database

Data availability

Data will be made available on request.

References

- Alesina, A., Harnoss, J., & Rapoport, H. (2016). Birthplace diversity and economic prosperity. *Journal of Economic Growth*, 21(2), 101–138.
- Anderson, J. E., & Van Wincoop, E. (2003). Gravity with gravitas: A solution to the border puzzle. *The American Economic Review*, 93(1), 170–192.
- Antonietti, R., & Franco, C. (2021). From FDI to economic complexity: A panel granger causality analysis. *Structural Change and Economic Dynamics*, 56, 225–239.
- Antràs, P., Chor, D., Fally, T., & Hillberry, R. (2012). Measuring the upstreamness of production and trade flows. *The American Economic Review*, 102(3), 412–416.
- Arrow, K. J. (1969). Classificatory notes on the production and transmission of technological knowledge. *The American Economic Review*, 59(2), 29–35.
- Bahar, D., Hausmann, R., & Hidalgo, C. A. (2014). Neighbors and the evolution of the comparative advantage of nations: Evidence of international knowledge diffusion? *Journal of International Economics*, 92(1), 111–123.
- Bahar, D., Rapoport, H., & Turati, R. (2022). Birthplace diversity and economic complexity: Cross-country evidence. *Research Policy*, 51(8), Article 103991.
- Baier, S. L., & Bergstrand, J. H. (2004). Economic determinants of free trade agreements. *Journal of International Economics*, 64(1), 29–63.
- Baldwin, R. (2013). Lessons from the European spaghetti bowl. *ADB working papers* 418. Asian Development Bank Institute.
- Batista, C., Lacuesta, A., & Vicente, P. (2007). *Brain drain or brain gain: Evidence from an African success story*. IZA Discussion Paper. No. 3035.
- Bitzer, J., & Kerekes, M. (2008). Does foreign direct investment transfer technology across borders? New evidence. *Economics Letters*, 100(3), 355–358.
- Blomström, M., & Kokko, A. (1998). Multinational corporations and spillovers. *Journal of Economic Survey*, 12, 247–277.
- Blomström, M., & Sjöholm, F. (1999). Technology transfer and spillovers: Does local participation with multinationals matter? *European Economic Review*, 43(4–6), 915–923.
- BLS. (2022). United States Bureau of statistics. <https://www.bls.gov/>.
- Boffa, M., Jansen, M., & Solleder, O. (2019). Do we need deeper trade agreements for GVCs or just a BIT? *The World Economy*, 42(6), 1713–1739.
- Borchert, I., Larch, M., Shikher, S., & Yotov, Y. V. (2022). Disaggregated gravity: Benchmark estimates and stylized facts from a new database. *Review of International Economics*, 30(1), 113–136.
- Bottazzi, L., & Peri, G. (2003). Innovation and spillovers in regions: Evidence from European patent data. *European Economic Review*, 47, 687–710.
- Braun, E. (2024). *Resolving the complexity puzzle: Economic complexity and positions in global value chains jointly explain economic development*. Group Economic Geography: Utrecht University, Department of Human Geography and Spatial Planning (No. 2401).
- Buckley, P. J., Clegg, J., & Wang, C. (2007). Is the relationship between inward FDI and spillover effects linear? An empirical examination of the case of China. *Journal of International Business Studies*, 38, 447–459.
- Campi, M., & Dueñas, M. (2019). Intellectual property rights, trade agreements, and international trade. *Research Policy*, 48(3), 531–545.
- Coe, D. T., & Helpman, E. (1995). International R&D spillovers. *European Economic Review*, 2921(94), 859–887.
- Coe, D. T., Helpman, E., & Hoffmaister, A. W. (2009). International R&D spillovers and institutions. *European Economic Review*, 53(7), 723–741.
- Commander, S., Chanda, R., Kangasniemi, M., & Winters, A. (2004). Must skilled migration be a brain drain? Evidence from the Indian software industry. *IZA Working Paper*. No. 1422.
- Deltas, G., Desmet, K., & Facchini, G. (2012). Hub-and-spoke free trade areas: Theory and evidence from Israel. *Canadian Journal of Economics*, 45(3), 942–977.

- Docquier, F., & Rapoport, H. (2007). Skilled migration: The perspective of developing countries. *Discussion Paper*, 17.
- Fan, F., Anwar, S., & Huang, S. (2018). Cultural diversity and export sophistication. *International Review of Economics & Finance*, 58, 508–522.
- Fan, Z., Zhou, Y., & Anwar, S. (2024). Centralization of trade agreements network and global value chain participation. *The Quarterly Review of Economics and Finance*, 94, 11–24.
- Felipe, J., Kumar, U., Abdon, A., & Bacate, M. (2012). Product complexity and economic development. *Structural Change and Economic Dynamics*, 23(1), 36–68.
- Findlay, R. (1978). Some aspects of technology transfer and direct foreign investment. *The American Economic Review*, 68(2), 275–279.
- Grossman, G. M. (2016). The purpose of trade agreements. *Handbook of Commercial Policy*, 1, 379–434.
- Hansen, M. T. (1999). The Search-transfer problem: The role of weak ties in sharing knowledge across organization subunits. *Administrative Science Quarterly*, 44(1), 82–111.
- Harding, T., & Javorcik, B. S. (2012). Foreign direct investment and export upgrading. *The Review of Economics and Statistics*, 94(4), 964–980.
- Hausmann, R., & Hidalgo, C. A. (2011). The network structure of economic output. *Journal of Economic Growth*, 16(4), 309–342.
- Hausmann, R., Hwang, J., & Rodrik, D. (2007). What you export matters. *Journal of Economic Growth*, 12(1), 1–25.
- Hidalgo, C. A., & Hausmann, R. (2009). The building blocks of economic complexity. *Proceedings of the National Academy of Sciences*, 106(26), 10570–10575.
- Hoang, D. P., Chu, L. K., & To, T. T. (2023). How do economic policy uncertainty, geopolitical risk, and natural resources rents affect economic complexity? Evidence from advanced and emerging market economies. *Resources Policy*, 85, Article 103856.
- Hofmann, C., Osnago, A., & Ruta, M. (2017). Horizontal depth: A new database on the content of preferential trade agreements. *World Bank Policy Research Working Paper*. No.7981.
- Hudson, J., & Minea, A. (2013). Innovation, intellectual property rights, and economic development: A unified empirical investigation. *World Development*, 46, 66–78.
- Hur, J., Alba, J. D., & Park, D. (2010). Effects of hub-and-spoke free trade agreements on trade: A panel data analysis. *World Development*, 38(8), 1105–1113.
- Jaffe, A. B., Trajtenberg, M., & Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *Quarterly Journal of Economics*, 108(3), 577.
- Johnson, R. C., & Noguera, G. (2017). A portrait of trade in value-added over four decades. *The Review of Economics and Statistics*, 99(5), 896–911.
- Keller, W. (2002). Geographic localization of international technology diffusion. *The American Economic Review*, 92(1), 120–142.
- Keller, W., & Yeaple, S. R. (2013). The gravity of knowledge. *The American Economic Review*, 103(4), 1414–1444.
- Krugman, P. R. (1991). The move toward free trade zones. *Economic review*, 76(6), 7–41.
- Laget, E., Osnago, A., & Rocha, N. (2020). Deep trade agreements and global value chains. *Review of Industrial Organization*, 57(2), 379–410.
- Laget, E., Osnago, A., Rocha, N., & Ruta, M. (2020). Deep trade agreements and global value chains. *Review of Industrial Organization*, 57, 379–410.
- Lee, W. (2019). Services liberalization and global value chain participation: New evidence for heterogeneous effects by income level and provisions. *Review of International Economics*, 27(3), 888–915.
- Lewbel, A. (2012). Using heteroscedasticity to identify and estimate mismeasured and endogenous regressor models. *Journal of Business & Economic Statistics*, 30(1), 67–80.
- Li, A. X. (2015). State–society synergy and export sophistication. *Economics & Politics*, 27(3), 433–458.
- Limão, N. (2016). Preferential trade agreements. *Handbook of commercial policy*, 1, 279–367.
- Manger, M. S., Pickup, M. A., & Snijders, T. A. B. (2012). A hierarchy of preferences: A longitudinal network analysis approach to pta formation. *Journal of Conflict Resolution*, 56(5), 853–878.
- Maurya, G., & Sahu, S. (2022). Cross-country variations in economic complexity: The role of individualism. *Economic Modelling*, 115, Article 105961.
- Moore, E. M., Dau, L. A., & Mingo, S. (2021). The effects of trade integration on formal and informal entrepreneurship: The moderating role of economic development. *Journal of International Business Studies*, 52(4), 746–772.
- Nelson, R. R., & Phelps, E. S. (1966). Investment in human, technological diffusion and economic growth. *The American Economic Review*, 56, 69–75.
- Neves, P. C., Afonso, O., Silva, D., & Sochirca, E. (2021). The link between intellectual property rights, innovation, and growth: A meta-analysis. *Economic Modelling*, 97, 196–209.
- Nguyen, C. P., Schinckus, C., & Su, T. D. (2020). The drivers of economic complexity: International evidence from financial development and patents. *International Economics*, 164, 140–150.
- Nunn, N., & Qian, N. (2014). US food aid and civil conflict. *The American Economic Review*, 104(6), 1630–1666.
- Orefice, G., & Rocha, N. (2014). Deep integration and production networks: An empirical analysis. *The World Economy*, 37(1), 106–136.
- Orlic, E., Hashi, I., & Hisarcikilar, M. (2018). Cross sectoral FDI spillovers and their impact on manufacturing productivity. *International Business Review*, 27(4), 777–796.
- Ornelas, E., Turner, J. L., & Bickwit, G. (2021). Preferential trade agreements and global sourcing. *Journal of International Economics*, 128, Article 103395.
- Pajek. (2022). *Pajek software, V. 5*. <https://en.freedownloadmanager.org/Windows-PC/Pajek-FREE.html>.
- Pauwelyn, J., & Alschner, W. (2015). Forget about the WTO: The network of relations between preferential trade agreements (PTAs) and “Double PTAs.”. In A. Dür, & M. Elsig (Eds.), *Trade cooperation: The purpose, design and effects of preferential trade agreements* (pp. 497–532). Cambridge: Cambridge University Press.
- Polanyi, M. (1962). *Personal knowledge: Towards a post-critical philosophy*. London: Routledge Press.
- Poncet, S., & de Waldemar, F. S. (2015). Product relatedness and firm exports in China. *The World Bank Economic Review*, 29(3), 579–605.
- Rivera-Batiz, L. A., & Romer, P. M. (1990). Economic integration and endogenous growth. *Quarterly Journal of Economics*, 106(2), 531–555.
- Ruta, M. (2018). Preferential trade agreements and global value chains: Theory, evidence, and open questions. *World Bank Policy Research Working Paper*. No. 8190.
- Sopranzetti, S. (2018). Overlapping free trade agreements and international trade: A network approach. *The World Economy*, 41(6), 1549–1566.
- Stojčić, N., Vojinić, P., & Aralica, Z. (2018). Trade liberalization and export transformation in new EU member states. *Structural Change and Economic Dynamics*, 47, 114–126.
- Vu, T. V. (2022). Does institutional quality foster economic complexity? The fundamental drivers of productive capabilities. *Empirical Economics*, 63(3), 1571–1604.
- Wang, J. (2024). Network of trade agreements, production capability and Economic Complexity. *Industrial Innovation*, 7, 5–7 (in Chinese).
- WDI. (2022). *World development indicators*. The World Bank. <https://databank.worldbank.org/source/world-development-indicators>.
- World Bank. (2023). *World Bank data*. <https://data.worldbank.org/>.
- Zeng, K., Lu, Y., & Li, Y. (2021). Trade agreements and global value chain participation: Evidence from Chinese industries. *Economics & Politics*, 33(3), 533–582.
- Zhang, R., Zhao, J., & Zhao, J. (2021). Effects of free trade agreements on global value chain trade: A research perspective of gvc backward linkage. *Applied Economics*, 53(44), 5122–5134.
- Zhou, Y., Fan, Z., & Anwar, S. (2023). Birthplace diversity of immigrants, capabilities and export structure transformation. *Economic Modelling*, 119, Article 106125.
- Zucker, L., & Darby, M. (2007). *Star scientists, innovation and regional and national migration*. NBER Working Paper, Article 13547.